Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 15: Things That Relate

Relations and Their Properties

Chartrand & Zhang, Section 5.1

Name: _____ Date: _____

Where we are

Chapter 2 gave you Cartesian products — all ordered pairs (a, b) with $a \in A$ and $b \in B$. Today: a **relation** is nothing more than a *subset* of one of those. Picking which pairs to keep is how you say that some things are related and others are not.

By the end of today you should be able to

1. State what a relation from A to B is.
2. Test a relation for reflexivity, symmetry, and transitivity.
3. Produce a counterexample when a property fails.

1. A relation is a set of pairs

Definition

A **relation** R from a set A to a set B is a subset of $A \times B$. When $B = A$ we call R a **relation on** A.
We write $a \mathrel{R} b$ to mean $(a, b) \in R$, and $a \not\mathrel{R} b$ otherwise.

So a relation is a subset of a Cartesian product — two ideas you already have, combined.

Example. $A = \{1, 2, 3\}$ and $R = \{(1, 1), (1, 2), (2, 3)\}$ is a relation on A. Then $1 \mathrel{R} 2$ is true and $2 \mathrel{R} 1$ is false.

How many relations are there?

A relation on A is a subset of $A \times A$. If $|A| = n$ then $|A \times A| = n^2$, so the number of relations on A is 2^{n^2}.
For $|A| = 3$ that is $2^9 = 512$. The counting is Lecture 11's power-set theorem applied to $A \times A$ — and it is 2^n turning up yet again.

Relations you already use. $=$ on $\mathbb{Z}$. $\leq$ on $\mathbb{R}$. $\subseteq$ on $\mathcal{P}(S)$. $\equiv$ on compound statements, from Week 1. "divides" on $\mathbb{N}$.

2. Three properties

The three properties

Let R be a relation on a set A.

Reflexive: $\forall a \in A,\ a\ R\ a$

Symmetric: $\forall a, b \in A,\ \big(a\ R\ b \Rightarrow b\ R\ a\big)$

Transitive: $\forall a, b, c \in A,\ \big((a\ R\ b \wedge b\ R\ c) \Rightarrow a\ R\ c\big)$

Every one is a universal statement — so Week 4 applies

To **prove** a property: take arbitrary elements and argue.
To **disprove** one: exhibit *specific* elements where it fails. For symmetry that means naming a and b with $a\ R\ b$ but $b \not R\ a$. For transitivity, name all three of a, b, c.
"It's not transitive" with no witnesses is not an answer.

3. Testing relations on $\mathbb{Z}$

$a\ R\ b$ **if...**	**Refl**	**Sym**	**Trans**	**Witness for a failure**
$a = b$	Y	Y	Y	—
$a \leq b$	Y	N	Y	$1 \leq 2$ but $2 \not\leq 1$
$a < b$	N	N	Y	$1 \not< 1$; $1 < 2$ but $2 \not< 1$
$a - b$ is even	Y	Y	Y	—

4. Two relations that fail in instructive ways

Relation R_1 on $\mathbb{Z}$: $a\ R_1\ b$ **if** $ab \geq 0$

Reflexive? Yes — $a \cdot a = a^2 \geq 0$ always.

Symmetric? Yes — $ab = ba$.

Transitive? No. Witness: $a = -1$, $b = 0$, $c = 1$. Then $(-1)(0) = 0 \geq 0$ and $(0)(1) = 0 \geq 0$, but $(-1)(1) = -1$, which is not ≥ 0.

Relation R_2 on $\mathbb{Z}$: $a\ R_2\ b$ **if** $ab > 0$

Reflexive? No. Witness: $a = 0$, since $0 \cdot 0 = 0 \not> 0$.

Symmetric? Yes. Transitive? Yes.

Zero is where these relations break

R_1 has all the properties except transitivity; R_2 has all except reflexivity. Changing $\geq$ to $>$ moved the failure from one property to another, and **both failures are caused by the single element** 0.

Two lessons. **The properties are independent** — having two says nothing about the third. And when hunting for a counterexample, **try the edge cases first**: zero, equal elements, the empty set.

5. Relations given by a list of pairs

$A = \{1, 2, 3\}$. For each, mark the properties it has.

Relation	R	S	T	Why it fails
$\{(1,1),(2,2),(3,3)\}$	Y	Y	Y	—
$\{(1,1),(2,2),(3,3),(1,2),(2,1)\}$	Y	Y	Y	—
$\{(1,1),(1,2),(2,3)\}$	N	N	N	$2 \not R 2$; $(1,2)$; $(1,2,3)$
$\{(1,2),(2,1)\}$	N	Y	N	$1 \not R 1$; $(1,2,1)$ needs $1R1$

A checking trick for small relations

Reflexive: every (a, a) must be present. Scan the diagonal.
Symmetric: every pair must appear with its reverse. Scan for orphans.
Transitive: for each chain (a, b) and (b, c), check (a, c) is present. The last row above fails because $(1, 2)$ and $(2, 1)$ force $(1, 1)$, which is missing.

Exit ticket

1. $A = \{1, 2, 3\}$, $R = \{(1,1),(2,2),(1,2)\}$. Which properties does R have?

2. Give a witness showing $<$ on $\mathbb{Z}$ is not symmetric.

Before Wednesday

Read Section 5.2, "Equivalence Relations," pp. 152–155.

Self-check (not collected). Section 5.1, Exercises 7, 9, and 11 — odd-numbered, answers in the back.

Homework 6 is due Friday, September 25, 11:59 p.m.

Sections 2.2 and 2.4. Problems 5 and 6 are proofs — write them in complete sentences.

Questions I still have